CLASSIFICATION OF COMPUTERS

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- The type of input that computer accepts, decides its nature and capabilities and the type of computation that it will be able to perform.
- some of the computers accept input in the form of signals.
- On the other hand, majority of computers take input in the form of digits. Some computers accept input in both the forms i.e. in the form of digits and signals.
- On the basis of input type, computers have been classified into following three categories:
  - 1. Analog computers
  - 2. Digital Computers
  - 3. Hybrid computers

Analog Computers

- Signals can be defined as continuous flow of physical identity. Temperature, pressure, water level, speed, voltage, current etc. are few examples of signals.
- Those computers, which operate upon signals (by accepting temperature, pressure, speed etc. as input) are called analog computers. Speedometer, thermometer, barometer, voltmeter, ammeter etc. are few examples of analog computers.

Digital Computers

- In computer terminology, alphabets, numbers, special characters etc. are called digits. Those computers, which accept digits (like name, age, e mail address etc.) as input, are called digital computers. Computers that we see in offices, shops, hotels etc. are digital computers.

Hybrid Computers

- Those computers, which accept both types of inputs i.e. signals and digits are called hybrid computers. • For example, computers placed in ICU (Intensive Care Unit) are examples of hybrid computers. They accept heartbeats of the patient (signals) as input. They also accept instructions that need to be carried out (digits). Similarly computers placed in atomic reactors (taking radioactivity level and instructions as input) and mines (which take carbon monoxide level and instructions as input) are other examples of hybrid computers.

TYPES OF COMPUTERS

- Digital computers can be classified into various categories, on the basis of their sizes. When we say, size of the computer, it refers to multiple factors like, size of the memory, number of terminals that can be connected to the computer, storage capacity of the hard disk, type of processor used in the computer etc. Digital computers are generally classified into following categories:
- Microcomputers
- Mini computers
- Mainframe computers
- Supercomputers

Micro Computers

- Microcomputer is the smallest category of computers, in which single microchip is used for two basic units i.e. ALU and Control Unit. This microchip is often referred to as microprocessor. Microcomputers are further classified into following categories:
- Personal computers
- Laptop computers

- Tablet PC
- Personal Digital Assistant (PDA)

Personal Computers (PC)

- Computers that you commonly see these days are examples of Personal Computers. In short, they are called PC. Their outward appearance is shown in the diagram next.
- They mainly comprise of four units i.e. CPU box called System Unit, keyboard, Monitor and mouse. Only one person can work on them at a time, so they are called single user computer.



Laptop Computers

- Laptop computers also fall into the category of microcomputers.
- They possess same capabilities and resources that of Personal Computers.



Tablet PCs

- Tablet PC is the latest development in the field of computers.
- It looks like laptop computer with many more enhancements and additional facilities packed, within same space.
- Display system of Tablet PC, vertically rotates, by 1800. When rotated, it starts accepting input through a special light pen, called light pen. Whatever is written on Tablet PC's screen, with the pen, in natural handwriting, is taken as input.

Personal Digital Assistants (PDA)

- Personal Digital Assistants are very small computers

that fit within a palm.

- They are often of the size of scientific calculator. This is the reason, why they are also called Palmtop computers. They are mostly used for storing addresses, telephone numbers, maintaining personal appointments etc.



Minicomputers

- Minicomputers are relatively larger and faster computers. Apart from single user environment, they also support multi user environment.
- Main characteristics of minicomputers are;
- They are built, using high performance and high capability processors.
- Memory size in such type of computers is generally very large.
- They support multiple terminals, which may range from 2 to 16.

Mainframes

•Mainframes fall into next higher category of computers. Their internal architecture and circuitry remains different from minicomputers. They use specially designed proprietary circuits instead of just single microprocessor as their CPU.

•Due to proprietary circuitry, mainframes become quite big in size and provide facility for connecting minicomputers and microcomputers with them. Few characteristics of mainframes are listed as follows;

• They possess the capability of addressing larger memory • sizes than that of minicomputers.

• They also support larger capacity disks like optical disks etc.

• Their terminal connectivity can go as high as 256 or • beyond.

• They possess large number of application libraries that • provide great help to developers in developing useful • applications.

Supercomputers

• Supercomputer is the largest category of computers. They use multiple CPUs for processing the data and executing the instructions.

• While performing a task through Supercomputer, the complete task gets divided into

multiple independent tasks.

- Each CPU takes up individual task and completes it in parallel. This methodology of processing is called parallel processing.